

Camera Gimbal Control from UAV Autopilot Target Tracking *

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ABSTRACT

Unmanned Aerial Vehicle (UAV) is better and safer to carry missions for aerial photography for terrain feature documentation and surveillance. An aerial photography mechanism for camera gimbal control in conjunction with UAV autopilot system is proposed in this paper. A GPS-to-GPS target tracking is implemented into autopilot flight control system to aim at the assigned fixed point target. In order to properly control dynamic photography, coordinate conversion from airborne gimbal frame into the Earth frame via UAV body frame coordinates should be correctly aligned to the target. To avoid gimbal lock problem, the 3D quaternion-based attitude estimation from GPS and IMU is formulated. The UAV autopilot and target tracking control can be implemented on microcontroller with a reliable and stable performance. A UAV autopilot is designed and implemented mounting with 3D gimbal fixture for realization and flight tests. Flight experiment results show good system performance to accomplish the proposed objective.

Keywords: UAV, Camera Gimbal Control, Coordinate Transformation, Flight Control

I. INTRODUCTION

Aerial photography brings in great impressions to the public because of its better views from bird eyes. Beyond the beautiful scenes, periodic terrain feature monitoring and recording via aerial photography is very useful for livelihood applications in environmental conservation as well as disaster monitoring from the air. To improve the photographic efficiency with less cost and better safety, it is feasible to use the Unmanned Aerial Vehicle (UAV) to execute the aerial mission remotely and automatically. By carrying camera as payloads, UAV can take aerial photography under any assigned missions, even risky and dangerous situations. UAV photography costs much less than manned operation on aircraft, and more efficient to finish jobs when weather permits. Eisenbeiß analyzed UAV aerial photography with advantages, restrictions and applications [1] to explore its usefulness and value. It brought UAVs in wider and useful applications. However it is not a simple task to carry on due to

complex problems to deal with.

In UAV aerial photography, real time GPS data will be acquired and used to match the mission points by automatic target-tracking [2]. UAV performance is generally executed by autopilot flight control with a designate waypoint-based flight path planning to the assigned targets [3]. A 3 degree-of-freedom (DOF) camera gimbal is selected to mount onto UAV to aim at the target point simultaneously with appropriate control when UAV is auto-piloted to fly into target area. A precise camera gimbal control is important in conjunction with flight control to track targets dynamically.

This paper presents UAV camera gimbal control for target alignment and photography by autopilot flight control. The key for camera to aim at the ground target relies on both UAV and target being identified by their GPS positions. A GPS-to-GPS alignment mechanism is adopted for target searching and aiming. For precisely aiming at the mission target, it requires that the camera gimbal frame, UAV body frame and the Earth frame

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should be conformed into one unique expression.

A light weight 3-axis gimbal system is selected from mature product for UAV with minor modifications. It is an off-the-shelf product and is easy to purchase and install. The camera gimbal should be mounted onto UAV near its center of gravity (CG). The coordinates shift from UAV CG should be measured for precise alignment calculation. To stabilize camera performance, UAV attitude estimation and stabilization is the most important part in autopilot flight control. An Inertia Measurement Unit (IMU) combining gyroscope, accelerometer, and sometimes with magnetometer [4, 5] is used to attitude identification on UAV frame. Based on the Inertia Measurement Unit (IMU) information, Attitude and Heading Reference System (AHRS) calculates and provides attitude data for flight control [6] and gimbal control. In order to calculate attitude more efficiently, quaternion matrix is applied to substitute Euler angle. Quaternions can solve the gimbal lock problem that occurs in Euler angle, and reduce the complication of rotation matrices [7]. To acquire the precise attitude data with better quality, various kinds of filters were proposed to integrate different sensor information and make AHRS work effectively [8]. Kalman filter is generally used to solve the linear problems, and can help to deal with the quaternion-based attitude estimation [9, 10]. In this study, an Extended Kalman filter is used to handle the nonlinear problems [11]. A Complementary filter [12] depends on different estimation feature is additionally used to adjust the weight constant of estimations and improve the final solution.

System integration of camera gimbal control for target tracking from UAV autopilot flight control system is accomplished in this study. Mathematical calculation by quaternion expression is formulated and programmed. The aerial photography using Ce-73 UAV with waypoint navigation and autopilot flight control is carried out by flight tests. Some preliminary results have demonstrated the effective operation of UAV photography.

II. FLIGHT SYSTEM ATTITUDE CONTROL

In order to implement target tracking for aerial photography, it is necessary to get UAV body frame to control the camera gimbal frame in correct coordinate conversion. By measuring the UAV attitude from IMU, an MCU (microprocessor control unit) is introduced to calculate the AHRS of roll, pitch and yaw.

In addition, the GPS receiver can provide vehicle's altitude, position, and even the heading. It combines IMU and GPS to construct the relationship between aircraft and target. Based on this relationship calculation, the MCU can output the corresponding PWM signal to control the camera gimbal to aim at the target. The UAV mission for aerial photography can be successfully accomplished.

AHRS consists of MEMS (Micro-Electro-Mechanical System) gyroscope, accelerometer, and magnetometer on three axes. For the difference between IMU and AHRS, IMU just provides

the sensor data, while AHRS offers additional on-board processing system to calculate data for attitude control. However, the data from IMU sometimes contain bias to influent AHRS solution. Therefore, initial calibration process is necessary to clear the bias. The UAV body frame is defined in Figure 1 for its coordinates of local (East, North, Up).

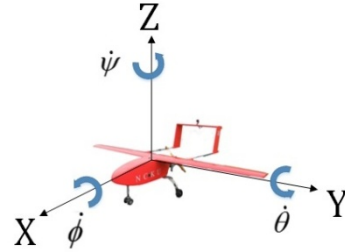


Figure 1 Definition of the body frame on Ce-73

From the gyroscope, the bias can be settled by setting its body frame in stationary condition for a little while. The error will be compensated to make the measurement of gyroscope $[\dot{\phi} \ \dot{\theta} \ \dot{\psi}]$ equal to $[0 \ 0 \ 0]$. The accelerometer likewise makes $[\ddot{\phi} \ \ddot{\theta} \ \ddot{\psi}]$ equal to $[0 \ 0 \ 1]$. The unit of accelerometer is referring to Earth gravity. However, the magnetometer is difficult to calibrate without standard value to compare with.

2.1 Attitude based on Euler Angle

Based on Figure 1, the corresponding Euler angle of rotation is defined and shown by the body axes frame about OX, OY, OZ of roll angle ϕ , pitch angle θ , yaw angle ψ , in counter-clockwise direction as positive. The rotate matrix is defines as:

$$\begin{aligned} R_x(\phi) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \\ R_y(\theta) &= \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \\ R_z(\psi) &= \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned} \quad (1)$$

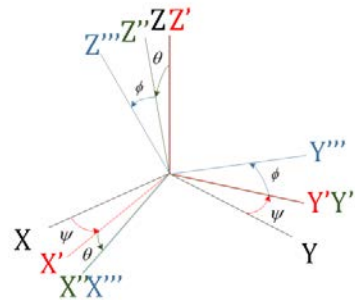


Figure 2 Coordinate Rotation

The series of rotations shown in Figure 2 follow the steps as:

1. Counter-clockwise rotation in the horizontal plane (Z-axis), through the yaw angle ψ , the *red frame*.
2. Counter-clockwise rotation about the pitch axis (Y-axis), through the pitch angle θ , the *green*

frame.

3. Counter-clockwise rotation about the roll axis (X-axis), through the roll angle ϕ , the *blue frame*.

After rotations, the rotation matrix becomes:

$$R_{xyz}(\phi, \theta, \psi) = R_x(\phi)R_y(\theta)R_z(\psi) = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \sin\phi\cos\theta \\ \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi & \cos\phi\cos\theta \end{bmatrix} \quad (2)$$

For convenience, the equation can be described as:

$$R_{xyz}(\phi, \theta, \psi) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad (3)$$

The inverse mapping makes the Euler angles in function of the rotation matrix as follows.

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} = \begin{bmatrix} \tan^{-1}\left(\frac{r_{23}}{r_{33}}\right) \\ -\sin^{-1}(r_{13}) \\ \tan^{-1}\left(\frac{r_{12}}{r_{11}}\right) \end{bmatrix} \quad (4)$$

Accordingly, the attitude of aircraft can be calculated by the trigonometric functions. However, Euler angle method has two problems to implement into MCU. First, the ‘‘Gimbal Lock’’ problem can happen when it losses one degree of freedom from real time data acquisition, and usually results in two of the three gimbals being driven into a parallel configuration. Second, this calculation includes too many trigonometric functions that brings heavy computation load to MCU. In order to avoid these two problems, the Quaternion angle is proposed to replace the Euler angle.

2.2 Attitude based on Quaternion

The quaternion is a 4×1 matrix whose element consists of a scalar part and a vector part. The scalar part is always real, and the vector part is always pure imaginary as [13]:

$$\hat{Q} = \begin{bmatrix} s \\ \vec{v} \end{bmatrix} = \begin{bmatrix} s \\ v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = q_0 + q_1\hat{i} + q_2\hat{j} + q_3\hat{k} \quad (5)$$

The quaternion product, denoted by $\otimes$, and the product relationship between $\hat{i}$, $\hat{j}$, $\hat{k}$ can be expressed by:

$$\begin{aligned} \hat{i}^2 = \hat{j}^2 = \hat{k}^2 &= -1 \\ \hat{i}\otimes\hat{j} = \hat{k} &= -\hat{j}\otimes\hat{i} \end{aligned} \quad (6)$$

$$\begin{aligned} \hat{j}\otimes\hat{k} = \hat{i} &= -\hat{k}\otimes\hat{j} \\ \hat{k}\otimes\hat{i} = \hat{j} &= -\hat{i}\otimes\hat{k} \end{aligned} \quad (7)$$

Quaternion is a number system containing wide and complex mathematical theorem [13]. In this paper for 3D coordinates transformation, only some computational rules are required for the attitude representation, such as the norm $\|\hat{Q}\|$, conjugate $\hat{Q}^*$, and inverse $\hat{Q}^{-1}$. In addition, to represent the attitude of a rigid body, the essential constraint should make its norm equal to unity. The computational rules are:

$$\begin{aligned} \|\hat{Q}\| &= \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1 \\ \hat{Q}^* &= q_0 - q_1\hat{i} - q_2\hat{j} - q_3\hat{k} \\ \hat{Q}^{-1} &= \frac{\hat{Q}^*}{\|\hat{Q}\|} = \hat{Q}^* \end{aligned} \quad (8)$$

According to the Euler’s rotational theorem, the quaternion can be defined by a rotational axis and a rotation angle, described as:

$$\hat{Q}(\theta) = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \cos\frac{\theta}{2} \\ \|\vec{e}\|\sin\frac{\theta}{2} \end{bmatrix} \quad (9)$$

where $\|\vec{e}\|$ is the normalized rotation axis and θ is the rotation angle.

Following the series of rotation, yaw, pitch and roll, the quaternion becomes:

$$\begin{aligned} \hat{Q}_{xyz}(\phi, \theta, \psi) &= \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \hat{Q}_z(\psi)\otimes\hat{Q}_y(\theta)\otimes\hat{Q}_x(\phi) \\ &= \left(\cos\frac{\psi}{2} + \sin\frac{\psi}{2}\hat{k}\right)\otimes\left(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\hat{j}\right)\otimes\left(\cos\frac{\phi}{2} + \sin\frac{\phi}{2}\hat{i}\right) \end{aligned} \quad (10)$$

$$\begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \cos \frac{\psi}{2} \cos \frac{\theta}{2} \cos \frac{\phi}{2} + \sin \frac{\psi}{2} \sin \frac{\theta}{2} \sin \frac{\phi}{2} \\ \cos \frac{\psi}{2} \cos \frac{\theta}{2} \sin \frac{\phi}{2} - \sin \frac{\psi}{2} \sin \frac{\theta}{2} \cos \frac{\phi}{2} \\ \cos \frac{\psi}{2} \sin \frac{\theta}{2} \cos \frac{\phi}{2} + \sin \frac{\psi}{2} \cos \frac{\theta}{2} \sin \frac{\phi}{2} \\ \sin \frac{\psi}{2} \cos \frac{\theta}{2} \cos \frac{\phi}{2} - \cos \frac{\psi}{2} \sin \frac{\theta}{2} \sin \frac{\phi}{2} \end{bmatrix} \otimes \begin{bmatrix} 1 \\ \hat{i} \\ \hat{j} \\ \hat{k} \end{bmatrix}$$

The quaternion rotation that rotates the vector $\vec{R}$ to $\vec{R}'$ satisfies:

$$\begin{aligned} \vec{R}' &= \hat{Q}^{-1} \otimes \vec{R} \otimes \hat{Q} = \hat{Q}^* \otimes \vec{R} \otimes \hat{Q} \\ \Rightarrow \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} &= \Omega_q \otimes \begin{bmatrix} x \\ y \\ z \end{bmatrix} \end{aligned} \quad (11)$$

where Ω_q is the rotation matrix of quaternion.

$$\begin{aligned} \Omega_q &= \begin{bmatrix} \Omega_{11} & \Omega_{12} & \Omega_{13} \\ \Omega_{21} & \Omega_{22} & \Omega_{23} \\ \Omega_{31} & \Omega_{32} & \Omega_{33} \end{bmatrix} \\ &= \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \end{aligned} \quad (12)$$

Substituting Equation (10) into Equation (12), the attitude angles can be calculated by:

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} = \begin{bmatrix} \tan^{-1}(\frac{\Omega_{23}}{\Omega_{33}}) \\ -\sin^{-1}(\frac{\Omega_{13}}{\Omega_{33}}) \\ \tan^{-1}(\frac{\Omega_{12}}{\Omega_{11}}) \end{bmatrix} \quad (13)$$

This equation can be compared with the Euler's form shown as Equation (5) by different notations.

2.3 Attitude update based on Quaternion

In order to update the quaternion attitude, it is necessary to solve the following differential equation:

$$\dot{\hat{Q}} = \frac{1}{2} \hat{Q} \otimes \hat{\omega} \quad (14)$$

where the $\hat{\omega}$ is the angular rate, which means $\hat{\omega} = 0 + \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$. With this definition, the classical Runge-Kutta integration algorithm applied.

After the update, the quaternion needs

normalization before it can be substituted into the rotation matrix Ω_q . With this equation, the attitude can be updated by the angular rates. However, since the gyroscope has “drift” naturally, it is not suitable for long estimation. The “complementary” data from others sensors is added for improvement. The accelerometer and magnetometer's measurements are used to calibrate the gyroscope data through complementary filter in real implementation. Following the calculations, the values of vectors, accelerations in both the Earth frame and body frame, are normalized. The measurements of magnetometer in the Earth frame contain horizontal and vertical components due to the inclination of the field, expressed by y axis, north and z axis. The vector of the magnetometer measurements of both the Earth frame and body frame can be obtained. The error of acceleration ΔA and the error of magnetic field ΔM can be calculated as:

$$\Delta A = \begin{bmatrix} 2(q_1q_3 - q_0q_2) - a_x \\ 2(q_2q_3 + q_0q_1) - a_y \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 - a_z \end{bmatrix} \quad (15)$$

$$\Delta M = \begin{bmatrix} 2m_{y,e}(q_1q_2 + q_0q_3) + 2m_{z,e}(q_1q_3 - q_0q_2) - m_x \\ m_{y,e}(q_0^2 - q_1^2 + q_2^2 - q_3^2) + 2m_{z,e}(q_2q_3 + q_0q_1) - m_y \\ 2m_{y,e}(q_2q_3 - q_0q_1) + m_{z,e}(q_0^2 - q_1^2 - q_2^2 + q_3^2) - m_z \end{bmatrix} \quad (16)$$

where $m_{x,e}$ $m_{y,e}$ $m_{z,e}$ are magnetometer measurements in the Earth frame, and m_x m_y m_z are those in body frame; a_x a_y a_z are accelerations of the body frame. In order to minimize the possible

errors from measurements, the body diagram of the attitude update with the complementary filter.

As ΔA and ΔM are normalized, Equation (14) can be modified into:

$$\begin{aligned}\dot{\hat{Q}} &= \frac{1}{2} \hat{Q} \otimes \hat{\omega} \Rightarrow \dot{\hat{Q}} = \frac{1}{2} \hat{Q} \otimes \hat{\omega}_c \\ \hat{\omega}_c &= (\hat{\omega} + k_p(\Delta A + \Delta M) + k_I \int (\Delta A + \Delta M))\end{aligned}\quad (17)$$

where the $\hat{\omega}_c$ is the calibrated angular rate. With the complementary filter, the attitude can be updated by the integrated information of gyroscope (ω), accelerometer (A), and the magnetometer (M). Therefore, the attitude estimation can maintain long term stability.

III. COORDINATE TRANSFORMATION

The coordinate transformation of aircraft and the tracking-target is the key importance to present. After the attitude estimation, the GPS information will be integrated into the relationship between the aircraft and tracking-target.

3.1 Relationship between UAV and Target

The position information can be expressed like:

$$P(\text{East}, \text{North}, \text{Altitude}) = P(x, y, z)$$

At the beginning, defining the direction of count-clockwise is positive, and the yaw angle ψ is zero to the north. Figure 3 describes the relationship between UAV and target.

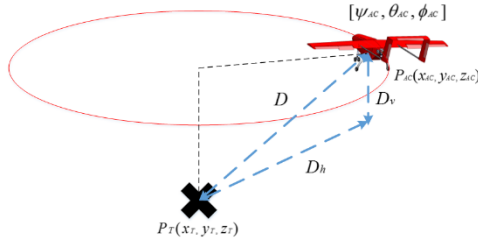


Figure 3 Relationship between UAV and target

The red ellipse is the flight route, ψ_{AC} , θ_{AC} , and ϕ_{AC} are yaw, pitch and roll of UAV P_{AC} to the target P_T . $P_T(x_T, y_T, z_T)$ and $P_{AC}(x_{AC}, y_{AC}, z_{AC})$ represent the position of target and the aircraft, respectively. The symbol D is the distance between the P_T and P_{AC} ; while the D_h and D_v mean the horizontal and vertical distance, respectively.

$$\begin{aligned}D &= \sqrt{(x_{AC} - x_T)^2 + (y_{AC} - y_T)^2 + (z_{AC} - z_T)^2} \\ D_h &= \sqrt{(x_{AC} - x_T)^2 + (y_{AC} - y_T)^2} \\ D_v &= \sqrt{(z_{AC} - z_T)^2}\end{aligned}\quad (18)$$

According to this relationship, ψ_T and θ_T , the aircraft yaw and pitch relative to the target can be calculated by:

$$\psi_T = \begin{cases} 360^\circ - \cos^{-1}\left(\frac{y_{AC} - y_T}{D_h}\right), & \text{if } x_{AC} - x_T > 0 \\ \cos^{-1}\left(\frac{y_{AC} - y_T}{D_h}\right), & \text{if } x_{AC} - x_T \leq 0 \end{cases}\quad (19)$$

$$\theta_T = \begin{cases} -\cos^{-1}\left(\frac{D_h}{D}\right), & \text{if } z_{AC} - z_T > 0 \\ \cos^{-1}\left(\frac{D_h}{D}\right), & \text{if } z_{AC} - z_T \leq 0 \end{cases}\quad (20)$$

where the angle range of ψ_T is 0° to 360° , and the θ_T is -90° to 90° . After these attitude angle being calculated, the attitude of camera gimbal can be defined as the ψ_{CG} , θ_{CG} , and ϕ_{CG} which are described in Figure 6 to be calculated by:

$$\psi_{CG} = 180^\circ + \psi_T - \psi_{AC}, \text{ and}$$

$$\begin{cases} \psi_{CG} - 360^\circ, & \text{if } \psi_{CG} > 360^\circ \\ \psi_{CG} + 360^\circ, & \text{if } \psi_{CG} < 0 \end{cases}\quad (21)$$

$$\theta_{CG} = -\theta_T - \theta_{AC}, \quad \phi_{CG} = -\phi_{AC}\quad (22)$$

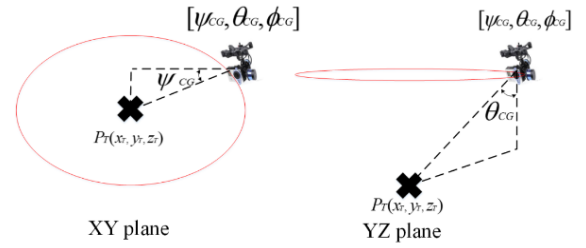


Figure 4 The attitude of camera gimbal

From the above equations and Figure 4, the desired gimbal attitude angles can be calculated.

IV. IMPLEMENTATION AND VERIFICATION

For the camera gimbal experiments, a UAV is selected using Ce-73 loading with autopilot system. Its flight route is an ellipse. The experiment setup of the auto-tracking aerial photography is presented with results and discussions.

4.1 Camera Gimbal Controller

Based on the IMU and GPS information, the MCU can calculate the aircraft attitude and the relative position between the aircraft and the target. Finally, MCU can be controlled based on the desired gimbal attitude angle with its corresponding PWM duty cycle value. The camera gimbal control system overview is shown in Figure 5.

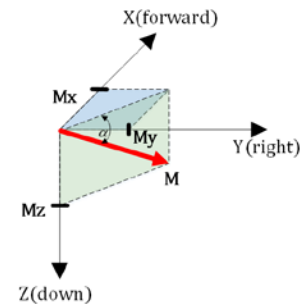


Figure 6 Magnetic field inclination angle

The angle α between the horizontal plane and the magnetic vector is called the inclination angle. The common way to estimate the heading is to measure and calculate the horizontal plane vector as:

$$\text{Heading } \psi = \arctan\left(\frac{H_y}{H_x}\right)$$

The magnetometer is used to read the Earth field. The output data distribute in circular shape by rotating the magnetometer 360° in the horizontal plane. However, the real measurement is always influenced and distorted by the environment. The original circle becomes an ellipse with the center shifted from the origin $[0, 0]$. To compensate the error, the scale factors and the offsets are applied to shift the center back to the origin $[0, 0]$. The simple calibration method is to make the magnetometer rotate in the horizontal plane and measure the minimum and maximum values for both X- and Y-axis.

4.2 Camera Gimbal Setup on UAV

In the implementation, MPU9150 is the original choice of IMU. MPU9150 is the integrated 9-axis motion tracking device that combines a 3-axis gyroscope, a 3-axis accelerometer and a 3-axis magnetometer. However, the magnetometer's data has received significant interference from the environment to devalue the accuracy of yaw angle. Therefore, HMC5983 is selected, which is a temperature compensated 3-axis magnetometers in order to obtain the accurate data of magnetometers.

The Ublox LEA-5H GPS receiver is selected to work with STM32 as MCU. The STM32 board is used to receive and calculate the attitude and GPS position. The MCU also needs to convert the desired gimbal attitude to the corresponding PWM signal to the gimbal controller.

Figure 7 is the camera gimbal controller, and the camera is Gopro Hero 3 with its mounting setup. This controller has PWM control mode and auto-stabilize mode. In the preliminary experiment, the roll angle selects with the auto-stabilize mode, and the yaw and pitch angle are used in PWM control mode.

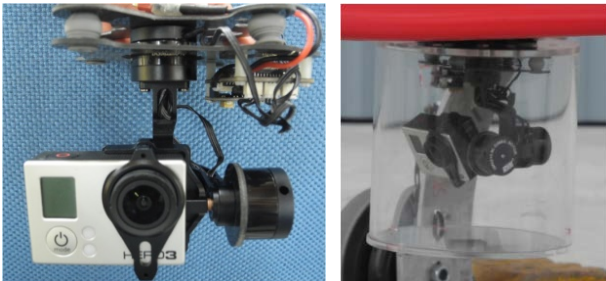


Figure 7 Camera Gimbal and its mounting

Frequencies of the information update, attitude calculation, and the control signal output are: IMU Update: >200 Hz, GPS Ublox Update: 5 Hz, AHRS Update: 100 Hz, PWM Update: 20 Hz.

Ce-73 UAV [15] shown in Figure 9 is installed a

Paparazzi autopilot system for preliminary tests with its specifications of full fiberglass and gasoline powered with Wing span: 3.6 m, Length: 2.5 m, Maxi. T/O Weight: 50 kg, Payload: 22 kg, Engine: 157cc, Fuel tank: 8 liters. The camera gimbal is setup under Ce-73 as shown in Figure 7 and Figure 8. A circular plastic shield is covered on the camera to reduce vibration due to wind.



Figure 8 Ce-73 with Paparazzi and Camera Gimbal

Paparazzi [16, 17] is an open source for free hardware and software project. The autopilot system hardware includes the flight control board, GPS receiver, and the wireless communication. UAV autopilot system is shown in Figure 9.

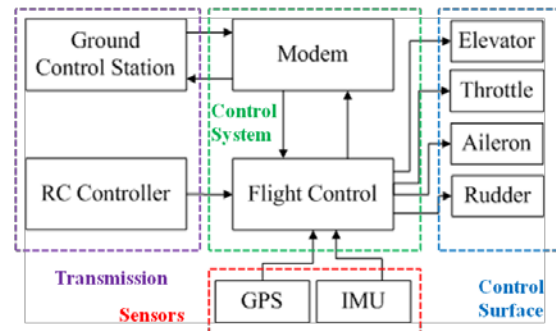


Figure 9 Scope View of UAV Autopilot System

4.3 Aerial Photography Auto Tracking

To verify the aerial photography, several targets assigned are assigned as waypoints for gimbal to track. The first target is selected at the crossroad near the JJ Airfield at 22.7724° N and 120.6232° E. The flight route is an ellipse above the crossroad.

Figure 10 describes the desired angle of yaw, pitch, and the flight route, where the blue line in the bottom figure is the flight route and the red circle presents as the crossroad. The red rectangles indicate the corresponding angle and position at photograph shootings. The pictures are set with screenshot by 5 second intervals. Since the gimbal system is an off-the-shelf product, it is difficult to get additional control response information from the mechanism.

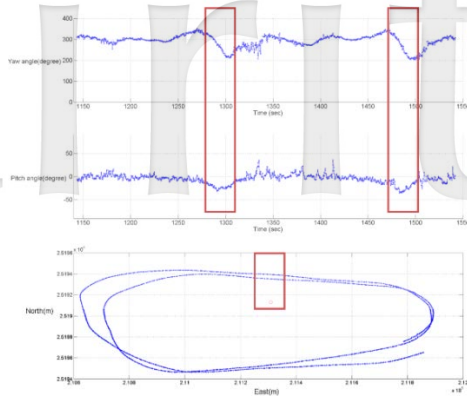


Figure 10 Desired angle (Yaw, Pitch) in flight route

Figure 11 shows the flight test results to aim at the same crossroad target by several pass-by flights, where a white house is significant to identify. Learning from the data, when the flight route is closer to the target, the change of angle is larger as shown in Figure 11. From the photographs, wind disturbance may result in minor shift to the target. Figure 12 shows other photography results to different waypoints.



Figure 11 Photography on the same target in three shootings

V. CONCLUSION

This paper develops the idea of UAV aerial photography from autopilot UAV. The relationship between UAV and the target can be constructed by the GPS-to-GPS matching and the trigonometric functions. To accurately control the camera gimbal, aircraft attitude should be acquired with appropriate coordinate transformation to aim at the target. To avoid gimbal lock and reduce computation load, the quaternion angle is proposed instead of the conventional Euler angle in coordinate transformation. Utilizing the IMU information, the quaternion theory for spatial coordinate expression is formulated. By applying a complementary filter to filter out the noises, an MCU can calculate the gimbal attitude on the aircraft promptly with precise camera control in auto-tracking mission.

To realize the proposed idea, UAV Ce-73 is loaded

for tests with Paparazzi autopilot system and executes the flight experiment of autopilot target tracking mission. Through the tests, the autopilot flight control and gimbal tracking can perfectly accomplish photo shootings during flights to acquire most pictures focusing on the target. This proves the gimbal control from UAV autopilot system is successful and accurate to accomplish the design objective. However, from the experiments, the strong wind affects mostly on the aiming alignment and causes to miss the target. The reliability and quality of aerial photography can further be improved by improving better and faster gimbal control response to compensate the wind disturbance. The success rate of the aerial photography can accomplish more than 95% from different weather conditions. The UAV autopilot target tracking for aerial photography is worthy of development.



Figure 12 Photography shootings on other waypoints

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